

Canadian Wildlife Service (CWS) analyses of the North American Breeding Bird Survey (1970-2024)

These are estimates of population trends and annual indices of relative abundance for years 1970 through 2024, derived from the North American Breeding Bird Survey (BBS). Population trends are estimates of the rate of change in the population size over a period of time expressed as an annual percent change (e.g., -1.5%/year between 1970-2024). Annual indices of relative abundance are estimates of the abundance of the species in each year, relative to all other years with data. The indices are expressed as an average expected count during a BBS survey in a given year and region. The series of annual indices for a given region describe the estimated population trajectory through time.

The BBS is an annual survey of birds conducted since 1966 in Canada and the United States. We provide trend estimates from 1970 onwards, to adjust for the changes to the number of surveys and regions covered by surveys during the first few years of the program. The surveys are conducted by volunteers and professionals who are highly skilled in bird identification, and who follow a rigorously designed survey protocol to record all the birds seen (within 400m) and all birds heard during 3-minute counts, at 50 pre-determined locations, along a roadside route. There are approximately 3000 of these routes surveyed in each year across Canada and the United States. The BBS is jointly coordinated by the Canadian Wildlife Service (Environment and Climate Change Canada) and the Eastern Ecological Science Center (United States Geological Survey). More information on the survey methods can be found online: <https://www.pwrc.usgs.gov/bbs/> or <https://www.canada.ca/en/environment-climate-change/services/bird-surveys/landbird/north-american-breeding.html>.

There are estimates here for >500 species of birds, and for all regions of Canada and the United States where there are sufficient data to estimate trends for each species. The estimates are available for the full survey area, Canada, United States, Canadian provinces and territories, US states, North American Bird Conservation Regions (BCRs), BCRs within each country (e.g., the Canadian portion of BCR 13), and for the geographic strata (spatial intersections of BCRs with the province/territory/state jurisdictions, such as BCR-13 in Ontario, Canada) that are used in the statistical analyses.

Any use of these BBS trend estimates should acknowledge the thousands of skilled volunteers in Canada and the United States who have participated in the BBS over the years and those who have served as national, provincial, territorial, or state coordinators.

These analyses use the 2025 release of the BBS data, which includes field data up to and including surveys conducted in 2024 (Ziolkowski, et al. 2025).

ANALYSIS

The model used to produce these estimates is a spatially explicit, hierarchical GAMYE model described in [Smith et al. 2024](#). The GAMYE structure estimates a medium- and long-term smooth population trajectory with annual fluctuations around the smooth. This spatial model is very similar to the GAMYE model described in [Smith and Edwards 2021](#), but shares information on abundance and trends among neighbouring geographic strata so that estimates of abundance and trends are better informed by local information (e.g., in situations where the data in a given stratum are relatively sparse, the model uses

information from the stratum's immediate neighbours to improve the estimates, rather than information from across the entire species' range).

The hierarchical structure of the non-linear model makes efficient use of the data and also means estimates are much less sensitive to random sampling error (e.g., variation in the number of routes run in a given year and region). The population trajectories in each geographic stratum are estimated in a way that shares some information across neighbouring regions in the species' range. Therefore, estimates of population trajectories in strata with relatively few data are more similar to the population trajectory in nearby regions. As a result, estimated rates of change in regions with relatively sparse data are generally more precise and less extreme than estimates from models that do not share information across the species' range.

The relative influence of local data and data from other parts of the species' range on the estimates of the population trajectory depends on two factors: 1) the quality of the data in the region, and 2) the relative shape of the local population trajectory, compared to the range-wide mean trajectory (i.e., range-wide means have greater influence when local estimates are very imprecise and/or are very different from the range-wide estimate). We believe this statistical assumption is reasonable because local populations of a given species are linked across the species' range, due to mixing of individuals through dispersal and migration. Also, many users of BBS trend estimates are likely comfortable with the idea of considering trend estimates from nearby regions to inform their understanding of a species' status in areas where local trend estimates are particularly poor (i.e., low precision) or are of an unexpected magnitude (e.g., increasing steeply when other regional trends suggest the species is stable or in decline). This is a justifiable feature of this new approach because: 1) populations of bird species are linked across political borders; 2) it improves the precision of local, regional, and national trend estimates; 3) for regions with high-quality data, it has little or no influence; 4) it reduces the likelihood of estimating anomalously extreme trends of very low quality (e.g., steeply increasing trends with exceptionally low precision); and 5) it allows us to estimate trends for more species.

Reliability measures

To quantify the reliability of the BBS trend estimates in describing overall population trends, three measures of reliability are calculated: precision, coverage, and local data weight. In addition to these three credibility measures, one may also want to consider the following measures reported in the final columns of the trend data tables: the number of routes used to generate the trends, the estimates of the probabilities that a given population has decreased or increased, or the probabilities that the magnitude of the population change falls within one of the six population change categories. These probabilities provide a concise and intuitive assessment of the uncertainty around some commonly desired interpretations. For example, a relatively strong positive trend that is imprecisely estimated may in fact have a greater probability of being a decline (i.e., the trend was estimated in the wrong direction), than a stable or slightly positive, but precisely estimated trend.

Precision: This measure reflects the width of the 95% credible interval (CI). Smaller values indicate a more precise trend estimate. The 95% CI defines a range of trend values that have a 95% probability of including the true population trend value, given the data and the model. For example, the following statement is a valid interpretation of a hypothetical 95% CI with lower limit = 1.2 and upper limit = 2.3:

given the BBS data and the accuracy of the model, there is a 95% probability that the average annual increase in the population is somewhere between 1.2%/year and 2.3%/year.

Geographic coverage: This measure indicates what proportion of the regional population had some probability of influencing the estimated trend. It is calculated as the estimated proportion of the regional population inside degree-blocks with BBS data. The reported value is a geometric average annual proportion covered during the time period of the trend, where degree blocks are considered covered for all years between the first and last years that contributing routes were run in the degree-block. Estimates of regional and degree-block specific population proportions were derived using maps of a species' relative density across its Canadian range. These maps were created by overlaying existing species range maps with estimates of relative population density within the species' range, which were derived using the average number of individuals on BBS routes run from 1997 through 2007, and comparable counts from other monitoring programs.

Local data weight: Indicates the proportion of the region and time series for which local data contributed to estimating the local population trajectory. That is, the proportion of the region with some data in each year, averaged across all years included in the trend. A value of 1.0 indicates that there are data from every year and stratum that makes up the region. Values less than 1.0 indicate that some portion of the region (i.e., one or more of the strata included in the region) does not include data from the full range of the time series, and therefore relies on the data from other regions in the species' range to estimate those years in the local trajectory.

Precision, coverage, and local data weight are combined into an overall reliability score for each trend. These overall reliability scores provide an overview that alerts the user to potentially unreliable trends. However, specific measures are also available to make an informed decision about the uncertainty associated with particular conclusions based on the trends. For example, a given trend may have a low reliability score because it is relatively imprecise (i.e., large credible interval), but if, for example, the magnitude of the trend estimate is sufficiently large, this imprecision may not affect an overall conclusion that the species' population has increased.

Changes to the spatial strata used in the 2024 analyses, relative to previous versions

Recently, some of the largest Bird Conservation Regions (BCR) in Northern Canada were changed (<https://www.birdscanada.org/bird-science/nabci-bird-conservation-regions>). These BCRs were subdivided to better reflect the latitudinal and longitudinal variation in bird communities across the Boreal forests. In addition, the boundaries between neighbouring BCRs in Eastern Canada were adjusted to more accurately represent the distributions of birds and their habitats. The spatial strata (smallest regions within-which BBS trends are analysed) are based on an intersection of BCRs with the political jurisdictions (Provinces, Territories, and States), so the changes to the BCR boundaries led to changes in the spatial strata, and therefore the grouping of BBS routes in the analyses and the area weights for each stratum (Figure 1). For example, subdividing the northern BCRs means there are more spatial strata and therefore fewer BBS routes in any given stratum, on average. In addition, some BBS routes are now associated with a different region because of the revised boundaries of BCRs. Finally, because the new BCR definitions have not yet been integrated into the main BBS database (Ziolkowski et al. 2025), BBS routes in this analysis were grouped into strata based on the spatial overlay of the route start locations.

with the updated BCR*State-Prov-Territory map. As a result, a small number of BBS routes that fall on or very close to the borders of BCRs may have been allocated to a different BCR than in previous analyses. Estimates for most species and regions are almost identical to previous versions. However, because of these revised BCRs and new spatial strata, estimates of population trends and the long-term population trajectories may have changed for a small number of species.

Note on the interim Bird Conservation Region numbers and codes. The new BCR formed by the subdivision of the large northern BCRs were given interim alpha-numeric codes when the boundaries were first defined, which were the codes used in these analyses. These interim alphanumeric codes were subsequently changed to numeric codes, which will be used in future analyses, and will be found on published maps. The alphanumeric codes correspond to the new numeric codes as follows: (3N = 68, 3C = 69, 3S = 70, 6N = 71, 6S = 72, 7W = 73, 7H = 74, 7E = 75, 8W = 76, 8E = 77). The English and French names of the BCRs remain the same as those used in the current analyses and can also be used to locate BCRs on the maps.

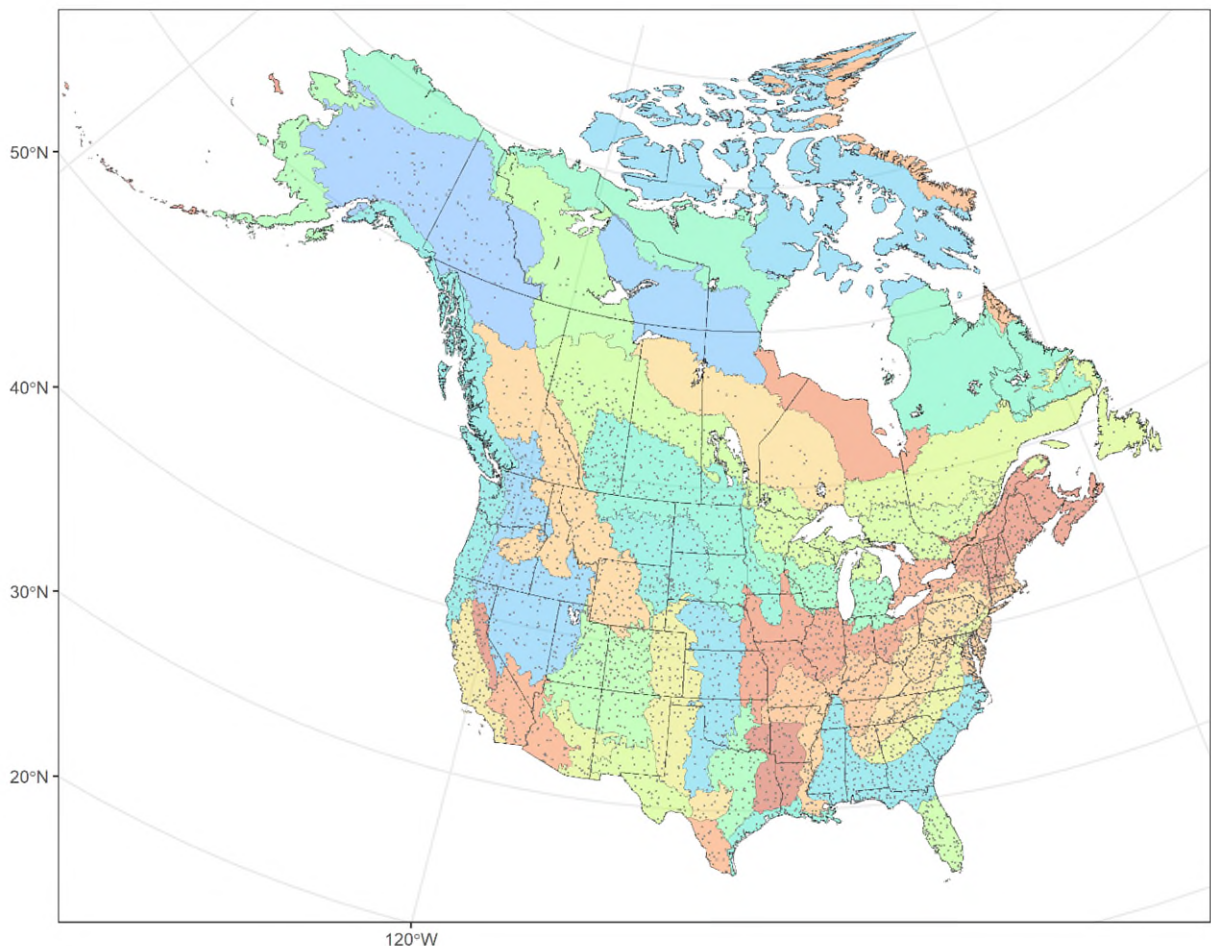


Figure 1. Analytical strata used in the analysis of North American Breeding Bird Survey data. Each region represents the spatial intersection of a political jurisdiction (province, state, territory, outlined in dark grey) with one of the North American Bird Conservation Regions (BCRs, coloured polygons with light grey outlines). The light grey points show the starting locations for all BBS routes that contributed data

to at least one species' trend model. Note: the most northerly strata (in Nunavut, Northern Quebec, and Northwest Territories) do not have sufficient BBS routes (> 2) to be included in any species' trend model.

Reproducibility

These estimates were produced using the functions and models in the R-package [bbsBayes2](#), with some additional code and custom functions to summarise the model output into the estimates here. To access the R-code used to analyse and summarize these estimates, visit

https://github.com/AdamCSmithCWS/BBS_Trends_CWS, or the permanent archive here:

<https://doi.org/10.5281/zenodo.20561464>. These repositories include all details and code to replicate (or customize) any of the analyses and results presented here.

For technical users, the raw model output files from the most recent annual version of these estimates are available by emailing the author and resource contact (see Contact Details, in the data catalogue record). With these raw output files, it is possible to generate trends (graphs, maps, etc.) for custom time-periods using [bbsBayes2](#) (e.g., if you want a trend estimate for the last 15-years in the western Boreal). The [documentation for bbsBayes2](#) includes examples of how to conduct these custom summaries.

CONTENTS

This includes the full set of estimated population trends, and annual indices of relative abundance for all years from 1970 through 2024 and for 521 species, from the Canadian Wildlife Service's analyses of the North American Breeding Bird Survey.

Results at a glance: BBS_trends_summary_2024.pdf

This file can be a useful way to scan through high-level results for each species. This file has one page per species that shows the continental and national trajectories (where applicable), as well as the long- and short-term trend maps, and trend estimates for national and continental scales.

Trends and Indices Files

There are several large .csv files that include **all** trend estimates and annual indices:

- All_Toutes_Tendance_BBS_Trends_2024.csv – all trend estimates for all species, regions, and time-periods (long-term, short-term, and three-generation)
- All_Toutes_BBS_Full_Indices_Complets_2024.csv – all full annual indices of relative abundance for all species, regions, and time-periods. These full annual indices include the annual fluctuations and are the most comprehensive display of the species' estimated population trajectories.
- All_Toutes_BBS_Smoothed_Indices_Lisses_2024.csv - all smoothed annual indices of relative abundance for all species, regions, and time-periods. These smoothed annual indices exclude the annual fluctuations and show only the smooth component of the species' population trajectories. These smoothed trajectories are used in estimating the trends (see Smith and Edwards, 2021 or Smith et al. 2024 for details).

These "All_" files are large. The annual indices files need to be queried using a program like R, because they have > 2 million rows and therefore exceed Excel's limits. The trend and index files have many

columns with a wealth of information, so there is an additional file that gives more detailed definitions of the information in each column “Column_explanation_explication_colonnes.xlsx”.

These large annual indices files have also been divided into smaller more manageable files based on the type of geographic region. There are subsets of the “All_BBS_” versions of the full indices and smoothed indices files that are small enough to be opened with Excel or other spreadsheet programs. Each one of these subset files includes estimates for a particular kind of regional breakdown:

- national, continental, and province/state estimates
- BCR and BCR-by-country
- Strata (the smallest geographic regions on which the analysis is stratified)

DISCLAIMER

The most recent version of these trend estimates are also available in an interactive web application through the Canadian Wildlife Service’s [BBS trend results website](#).

CITATIONS

The data citation for the raw BBS data:

Ziolkowski, D.J., M. Lutmerding, S.M. Skalos, W.B. English, and M-A.R. Hudson. 2025. North American Breeding Bird Survey Dataset 1966-2024: U.S. Geological Survey data release, <https://doi.org/10.5066/P14SNUV4>

REFERENCES

Smith, Adam C, Allison D Binley, Lindsay Daly, Brandon P M Edwards, Danielle Ethier, Barbara Frei, David Iles, Timothy D Meehan, Nicole L Michel, and Paul A Smith. 2024. Spatially Explicit Bayesian Hierarchical Models Improve Estimates of Avian Population Status and Trends. Ornithological Applications:duad056. <https://doi.org/10.1093/ornithapp/duad056>.

Smith, Adam C, and Brandon P M Edwards. 2021. North American Breeding Bird Survey Status and Trend Estimates to Inform a Wide Range of Conservation Needs, Using a Flexible Bayesian Hierarchical Generalized Additive Model. The Condor:duaa065. <https://doi.org/10.1093/ornithapp/duaa065>.